

# WRRRA-M 0.3 Source Law Filter Selection

Existence Uniqueness Convergence and Stability in the Four Filter Sector

|         |                               |
|---------|-------------------------------|
| Author  | Wonsik Choi                   |
| ORCID   | 0009-0001-4263-9772           |
| Email   | janefather@gmail.com          |
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| Status  | Conditional selection theorem |

## Research decision

This note takes the single open assumption left by WRRRA-M 0.2 and turns it into a finite selection problem. The cross-dimensional filter is not added as independent machinery. It is defined as the stable winner of a Source-supplied compatibility table under a fixed Law. The result is exact within a four-filter admissible class and remains conditional on the physical origin of the compatibility values.

## Abstract

WRRM 0.3 defines the minimal admissible sector of four one-to-one channel pairings that preserve the sixteen channels and the common color action inherited from version 0.2. Source supplies two scalar compatibility matrices, one for the left block and one for the right block. Law assigns each filter the sum of its selected compatibilities and evolves positive filter weights by a constant-score replicator flow. Two selection gaps completely decide the outcome. The 0.2 cross-dimensional filter is the unique maximizer exactly when both gaps are positive. In that case the flow converges to it from every interior initial condition, with an explicit exponential bound. The same filter survives entrywise score perturbations of size  $\rho$  whenever the smaller gap exceeds four  $\rho$ . This establishes existence, conditional uniqueness, convergence, and perturbation stability inside the stated four-filter sector. It does not derive the compatibility matrices from fundamental physics or show that nature gives the required positive gaps.

## Result in one sentence

Within the four-filter sector, Source and Law select the WRRM 0.2 cross filter uniquely and stably if and only if both independently fixed selection gaps are positive.

## Scope and stopping line

Version 0.3 addresses filter selection only. It does not change the sixteen-channel algebra of 0.2 and does not use the recovered Standard Model charges as inputs to the selection score.

### Included in 0.3

- A complete four-filter admissible class for the two binary pairing decisions
- Source compatibility matrices and a fixed scalar score
- A necessary and sufficient two-gap selection criterion
- A closed-form selection flow and convergence bound
- A worst-case perturbation stability margin
- A non-circular evaluation protocol and explicit falsifiers

### Reserved for 0.4 or later

- A first-principles calculation of the compatibility matrices
- Proof that the four-filter class is complete in a deeper theory
- Time-dependent scores, stochastic Source noise, or quantum amplitudes
- Three generations, masses, symmetry breaking, gravity, dark matter, cosmology, or consciousness

## Continuity from version 0.2

Version 0.2 fixed the input space and proved the consequences of one target pairing. It left the pairing itself unselected. Version 0.3 keeps that input and target unchanged, then compares it with every alternative obtained by independently swapping the left and right singlet assignments.

$$C_{16} = \mathbf{1}_0 \oplus \mathbf{7}_A \oplus \mathbf{7}_B \oplus \mathbf{1}_N$$

$$F_{\times} : C_{16} \longrightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\overline{\mathbf{4}}, \mathbf{1}, \mathbf{2})$$

The first filter label records the left assignment and the second records the right assignment. D means the direct left assignment or direct right assignment defined below. X means its swap. The target filter is F-DX: direct on the left and crossed on the right.

$$P_4 = \{F_{DX}, F_{XX}, F_{DD}, F_{XD}\}$$

## Fixed WRRR evaluation rule

| Stage                        | Question answered in this note                                                       |
|------------------------------|--------------------------------------------------------------------------------------|
| Verification input           | Which 0.2 channels and which established finite-flow identities are held fixed       |
| WRRR-specific transformation | How Source scores candidate pairings and how Law evolves their weights               |
| Output                       | Which gap signs select the cross filter and how fast the weight converges            |
| Falsification condition      | Which independently fixed score or dynamical failure rejects the 0.3 selection claim |

## The four admissible filters

The minimal candidate class changes only the two binary matchings that 0.2 needs. Every candidate is a channel-preserving bijection. No candidate mixes color components or changes the common SU(3) action. Once the labels A, B, N, and 0 are fixed, the four candidates are distinct.

| Filter | Left assignment               | Right assignment                        | Relation to 0.2     |
|--------|-------------------------------|-----------------------------------------|---------------------|
| F-DX   | 3-A with 1-A and 3-B with 1-B | anti-3-A with 1-N and anti-3-B with 1-0 | Target cross filter |
| F-XX   | 3-A with 1-B and 3-B with 1-A | anti-3-A with 1-N and anti-3-B with 1-0 | Left swap           |
| F-DD   | 3-A with 1-A and 3-B with 1-B | anti-3-A with 1-0 and anti-3-B with 1-N | Right swap          |
| F-XD   | 3-A with 1-B and 3-B with 1-A | anti-3-A with 1-0 and anti-3-B with 1-N | Both swaps          |

## Source compatibility values

Source supplies two real scalar tables. The left table compares each triplet origin with each internal singlet origin. The right table compares each antitriplet origin with the distinguished null and invariant singlets. These values are pairing compatibilities, not linear maps between inequivalent SU(3) representations. Their scalar form therefore does not mix or alter the color action.

$$W_L = (\ell_{ij})_{i,j \in \{A,B\}}, \quad W_R = (r_{ij})_{i \in \{A,B\}, j \in \{N,0\}}$$

The values may later be computed from residue, transition cost, or another upstream invariant. Version 0.3 does not choose that construction. It requires the tables to be fixed before the target filter is evaluated.

## Law and the filter score

Law gives each candidate the sum of the four compatibility entries selected by its two matchings. Additive constants common to every entry of one table do not affect the winner, and positive rescaling does not change a gap sign. The target score is shown below.

$$C(F_{DX}) = \ell_{AA} + \ell_{BB} + r_{AN} + r_{B0}$$

| Filter | Total score                               |
|--------|-------------------------------------------|
| F-DX   | $\ell_{AA} + \ell_{BB} + r_{AN} + r_{B0}$ |
| F-XX   | $\ell_{AB} + \ell_{BA} + r_{AN} + r_{B0}$ |
| F-DD   | $\ell_{AA} + \ell_{BB} + r_{A0} + r_{BN}$ |
| F-XD   | $\ell_{AB} + \ell_{BA} + r_{A0} + r_{BN}$ |

## Selection gap theorem

Define the left gap as the direct-left score minus the swapped-left score, and the right gap as the crossed-right score minus the direct-right score. Because the candidate set is finite and nonempty, at least one maximizing filter always exists. The target filter is the unique maximizer exactly when both gaps are positive.

$$\Delta_L = (\ell_{AA} + \ell_{BB}) - (\ell_{AB} + \ell_{BA}), \quad \Delta_R = (r_{AN} + r_{B0}) - (r_{A0} + r_{BN})$$

## Proof

Subtract each competing score from the target score. The three differences are the left gap, the right gap, and their sum. If both gaps are positive, all three differences are positive. Conversely, if the target is unique, it must beat the candidate that differs only on the left and the candidate that differs only on the right, so each gap must be positive. A zero gap produces at least a two-way tie. A negative gap selects the corresponding swapped assignment.

$$C_{DX} - C_{XX} = \Delta_L, \quad C_{DX} - C_{DD} = \Delta_R, \quad C_{DX} - C_{XD} = \Delta_L + \Delta_R$$

$$F_x = F_{DX} \text{ is unique} \iff \Delta_L > 0 \text{ and } \Delta_R > 0$$

| Gap condition               | Selection result                                               |
|-----------------------------|----------------------------------------------------------------|
| Delta-L > 0 and Delta-R > 0 | F-DX is unique                                                 |
| Delta-L = 0 or Delta-R = 0  | At least two candidates tie                                    |
| Delta-L < 0                 | The left-swapped candidate beats its direct-left counterpart   |
| Delta-R < 0                 | The right-direct candidate beats its crossed-right counterpart |

## Selection flow

To turn score ordering into a dynamical selection, Law evolves a positive weight  $p_F$  for each filter by the constant-score replicator equation. This standard flow preserves positivity and the unit sum [2]. Here it is used as a defined WRR selection law rather than as a biological analogy.

$$\dot{p}_F = \eta p_F (C(F) - \bar{C}), \quad \bar{C} = \sum_{G \in \mathcal{F}_4} p_G C(G)$$

For constant scores the flow has the closed solution below. The ratio of any competitor to the target decays exponentially at the corresponding score difference. If the target starts with nonzero weight and both gaps are positive, its weight approaches one.

$$p_F(t) = \frac{p_F(0)e^{\eta C(F)t}}{\sum_{G \in \mathcal{F}_4} p_G(0)e^{\eta C(G)t}}$$

The smallest competing score difference is delta, the smaller of the two gaps. It gives a uniform convergence bound and an explicit time sufficient to reach target weight at least one minus epsilon.

$$1 - p_{\times}(t) \leq \frac{1 - p_{\times}(0)}{p_{\times}(0)} e^{-\eta \delta t}, \quad \delta = \min(\Delta_L, \Delta_R)$$

$$t \geq \frac{1}{\eta \delta} \log\left(\frac{1 - p_{\times}(0)}{p_{\times}(0)\epsilon}\right) \implies p_{\times}(t) \geq 1 - \epsilon$$

The interior-start condition is necessary. A candidate with exactly zero initial weight remains absent under this flow. Version 0.3 therefore treats nonzero initial support for every admissible filter as part of Source.

## Perturbation stability

Suppose every compatibility entry changes by at most  $\rho$ . Each gap contains four signed entries, so its worst-case change is at most four  $\rho$ . The target selection therefore survives every such perturbation when the smaller original gap is greater than four  $\rho$ . This bound is sufficient and worst-case sharp for independent entry perturbations.

$$\|\widetilde{W} - W\|_{\infty} \leq \rho, \quad \min(\Delta_L, \Delta_R) > 4\rho \implies \widetilde{F}_* = F_{\times}$$

## Non circular evaluation protocol

The theorem becomes evidence for WRRM only under the following order.

1. Specify the admissible filter class without using the desired charge output.
2. Compute or measure both compatibility tables from Source and Law by a rule fixed in advance.
3. Freeze the tables, the rate  $\eta$ , and the initial support before checking the target.
4. Evaluate both gaps and publish their signs, magnitudes, and uncertainty bounds.
5. Apply the flow without retuning and compare the selected filter with F-DX.

Choosing compatibility values after seeing the 0.2 target is calibration to the answer, not an independent selection result. Until upstream values are supplied, 0.3 is an exact conditional theorem and a measurement protocol, not empirical confirmation.

## Claim ledger

| Code | Status                  | Statement in WRR-M 0.3                                                                                 |
|------|-------------------------|--------------------------------------------------------------------------------------------------------|
| E    | External input          | The 0.2 channel space and the standard constant-score replicator identity                              |
| D    | Architecture definition | The four-filter class, Source compatibility tables, additive score, and Law flow                       |
| T    | Exact theorem           | Two-gap uniqueness criterion, closed flow solution, convergence bound, and perturbation margin         |
| C    | Conditional result      | F-DX is uniquely and stably selected when both fixed gaps are positive                                 |
| H    | Unproved hypothesis     | Nature supplies the stated candidate class, positive gaps, and nonzero initial support                 |
| O    | Outside this version    | Upstream score physics, additional filters, generations, masses, gravity, cosmology, and consciousness |

## Falsification conditions

The 0.3 selection claim fails within its stated scope if any one of the following occurs.

- An independently specified Source and Law give Delta-L less than or equal to zero or Delta-R less than or equal to zero.
- An admissible fifth filter is found with a score at least as large as F-DX, invalidating completeness of the four-filter sector.
- The filter weights do not obey the published ratio law under fixed scores and rate.
- The observed score uncertainty is large enough that the claimed stability margin does not hold.
- Compatibility values or the admissible class are changed after the target outcome is inspected.

## Exact boundary

Version 0.3 derives no numerical compatibility entry from fundamental physics. It proves what follows once Source supplies independently fixed scalar compatibilities and Law uses the stated flow. Existence means existence of a maximizer in the finite admissible class, not proof that a physical pre-spacetime system realizes that class. Uniqueness and stability are conditional on positive measured or derived gaps.

## Conclusion

WRRM 0.2 showed what the cross-dimensional filter would produce. Version 0.3 states the minimal conditions under which Source and Law select that filter rather than one of its three pairing alternatives. Two numbers decide the result. Positive left and right gaps make the target unique, drive its weight to one at a calculable rate, and protect it against bounded score errors. Zero or negative gaps expose failure without interpretive escape. The remaining task is now narrower and empirical in form: define an upstream compatibility rule, freeze it, and determine the two gaps without using the desired particle assignment as an input.

## Appendix A Compact derivation

| Step | Fixed input                     | WRRM operation                      | Output                                    |
|------|---------------------------------|-------------------------------------|-------------------------------------------|
| 1    | The 0.2 sixteen-channel space   | Enumerate two binary matchings      | Four admissible filters                   |
| 2    | Two Source compatibility tables | Sum selected entries                | One score per filter                      |
| 3    | Four scores                     | Subtract competitors from F-DX      | Delta-L Delta-R and their sum             |
| 4    | Positive gaps                   | Apply constant-score selection flow | Convergence to F-DX                       |
| 5    | Entrywise error bound rho       | Propagate worst-case gap error      | Selection stable if min gap exceeds 4 rho |

$$\Delta_L = (\ell_{AA} + \ell_{BB}) - (\ell_{AB} + \ell_{BA}), \quad \Delta_R = (r_{AN} + r_{BO}) - (r_{AO} + r_{BN})$$

$$F_x = F_{DX} \text{ is unique} \iff \Delta_L > 0 \text{ and } \Delta_R > 0$$

## Appendix B Verification identities

The three score-difference coefficient identities and the normalization of the closed-form flow were checked with exact arithmetic during the document build. No floating-point approximation is used in the theorem.

$$p_F(t) = \frac{p_F(0)e^{\eta^{C(F)}t}}{\sum_{G \in \mathcal{F}_4} p_G(0)e^{\eta^{C(G)}t}}$$

## Appendix C References

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